

Stats → Random variable

eg. Rolling a dice → random experiment

→ $X = \{1, 2, 3, 4, 5, 6\}$

← 1 to 6

Random variable

$P(X=1) = \frac{1}{6}$ ✓

$P(X=H) = \frac{1}{2}$

tossing a coin

→ $Y = \{H, T\}$

$$P(X = \text{even}) = \frac{3}{6} \Rightarrow \frac{1}{2}$$

$$P(X=2) = \frac{1}{6}$$
$$X = \{1, 2, 3, 4, 5, 6\}$$

Discrete
D.R.V.

Random variable
begin Range

Continuous
Random variable

Real

$$Z = \{122.2, 146.4, 132\}$$
$$P(Z > 122) = ?$$

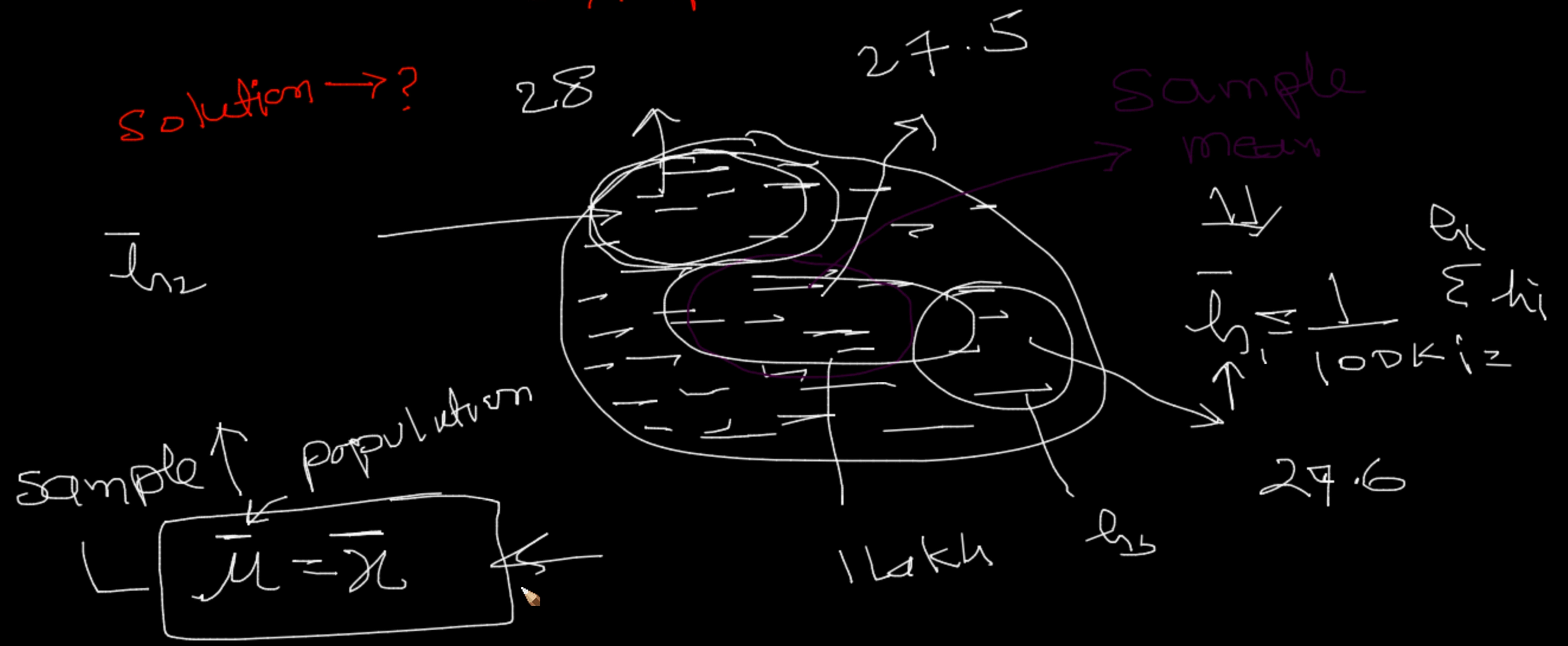
12.26, 157.23
butlers

⇒ Population and Sample

the mean height of human (μ) = $\frac{1}{N} \sum_{i=1}^N h_i$

↳ impossible x

solution → ?

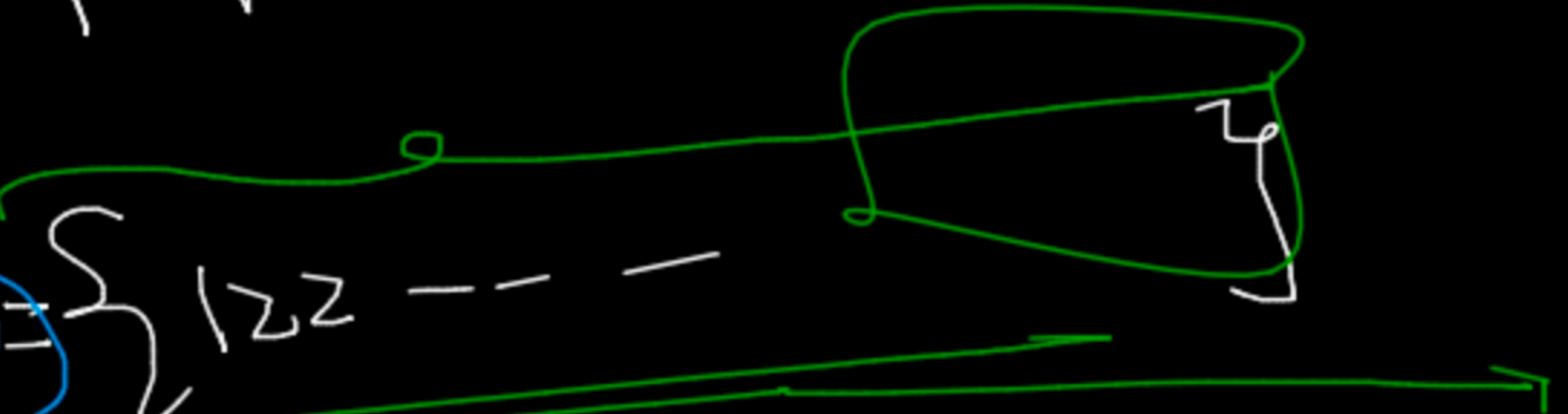


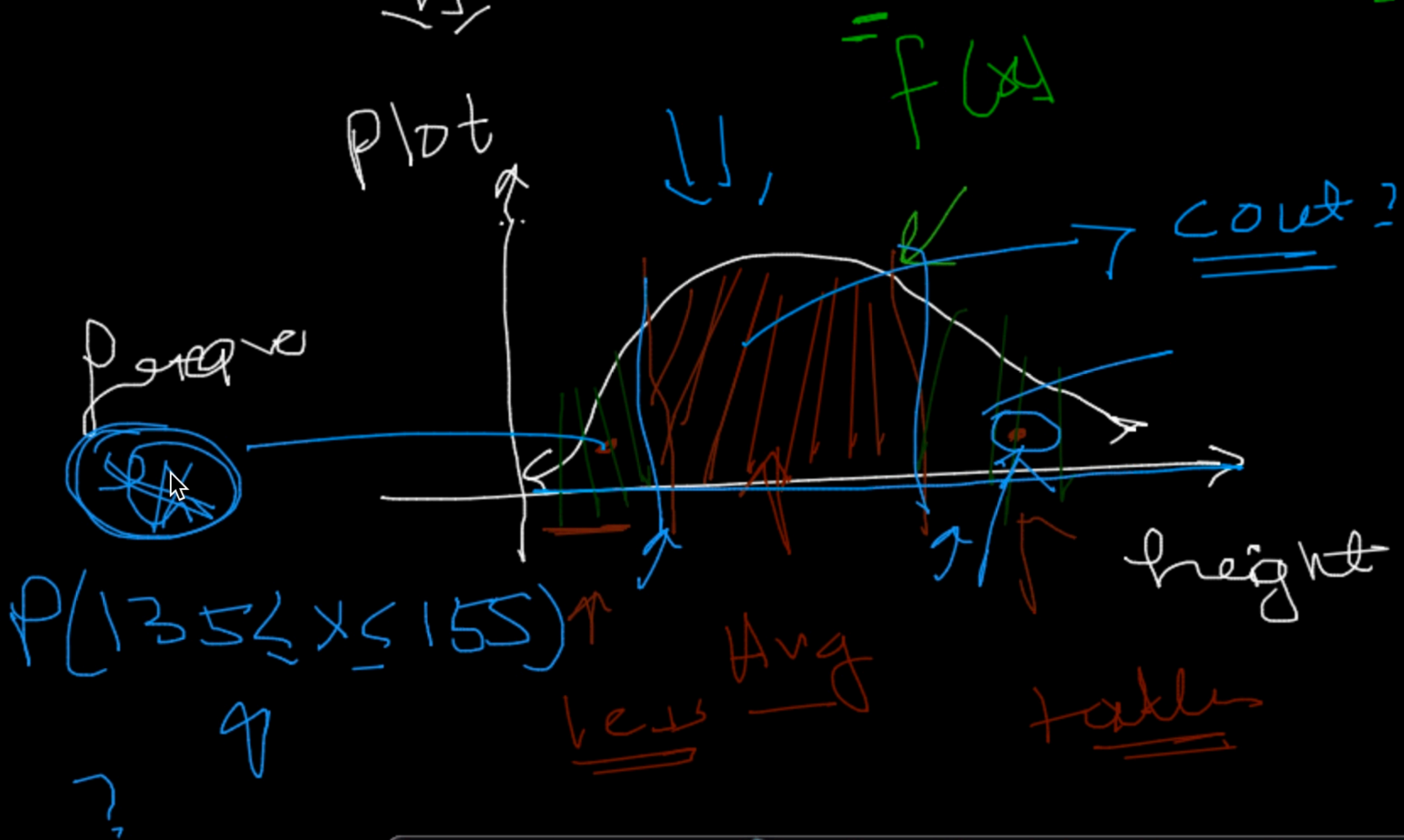
$$D_x \rightarrow [120, 121 \mid 122, 126, 127, 128, 129] \xrightarrow{\infty} \mathbb{N}$$

$\rightarrow \underbrace{p(l_n > 121)} = \frac{5}{7} \quad \boxed{\checkmark} \quad \boxed{\times}$

But?

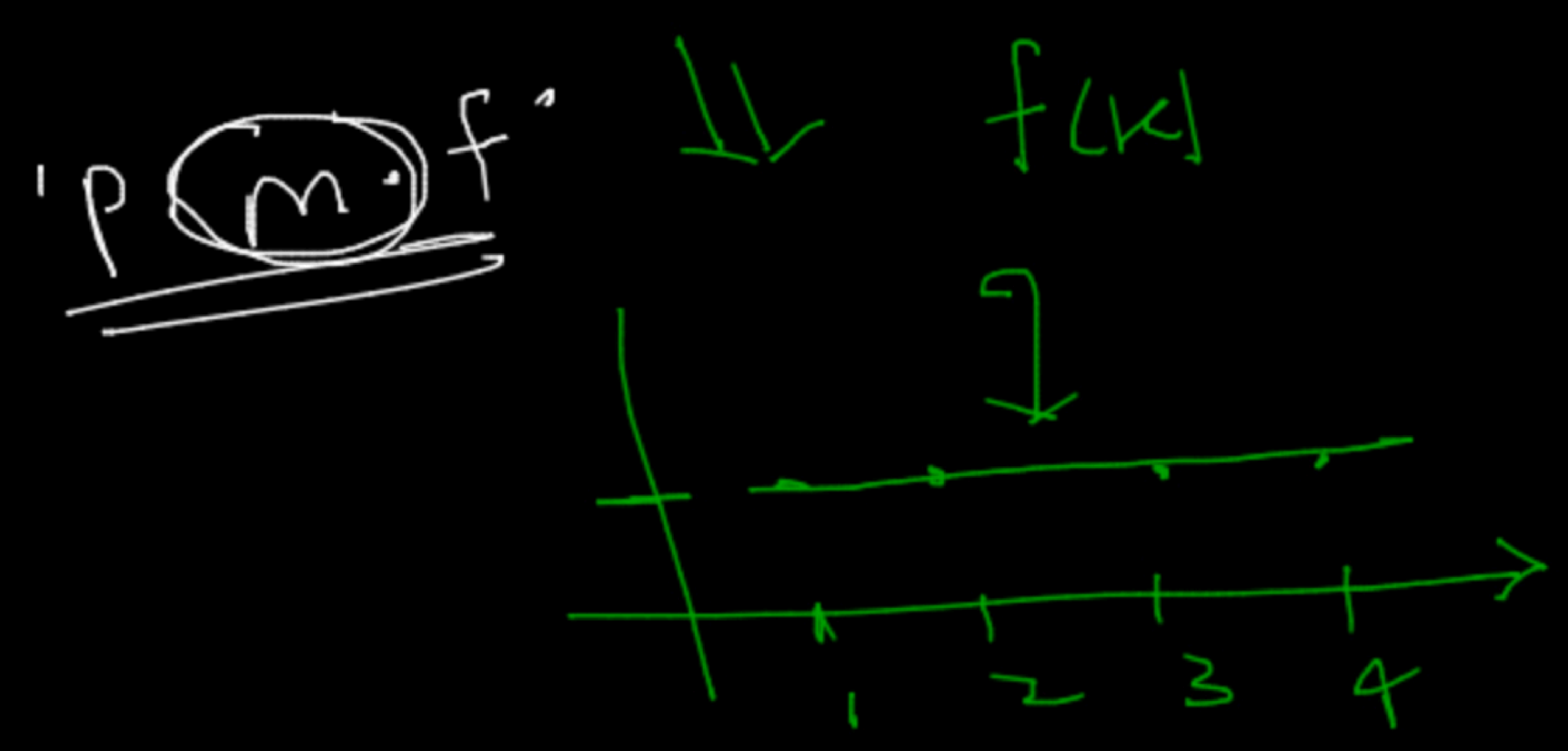
Gaussian Distribution

p.p.f
 $P_n = \sum_{i=1}^n p_i$  plot



D.R.V
 $X \rightarrow \{1, 2, 3, 4, 5, 6\}$
 $P(x) \quad 1/6 \quad 1/6$

$\sum P(x) = 1$
(0-1)



h_i
120
125
130
135
145
⋮
120

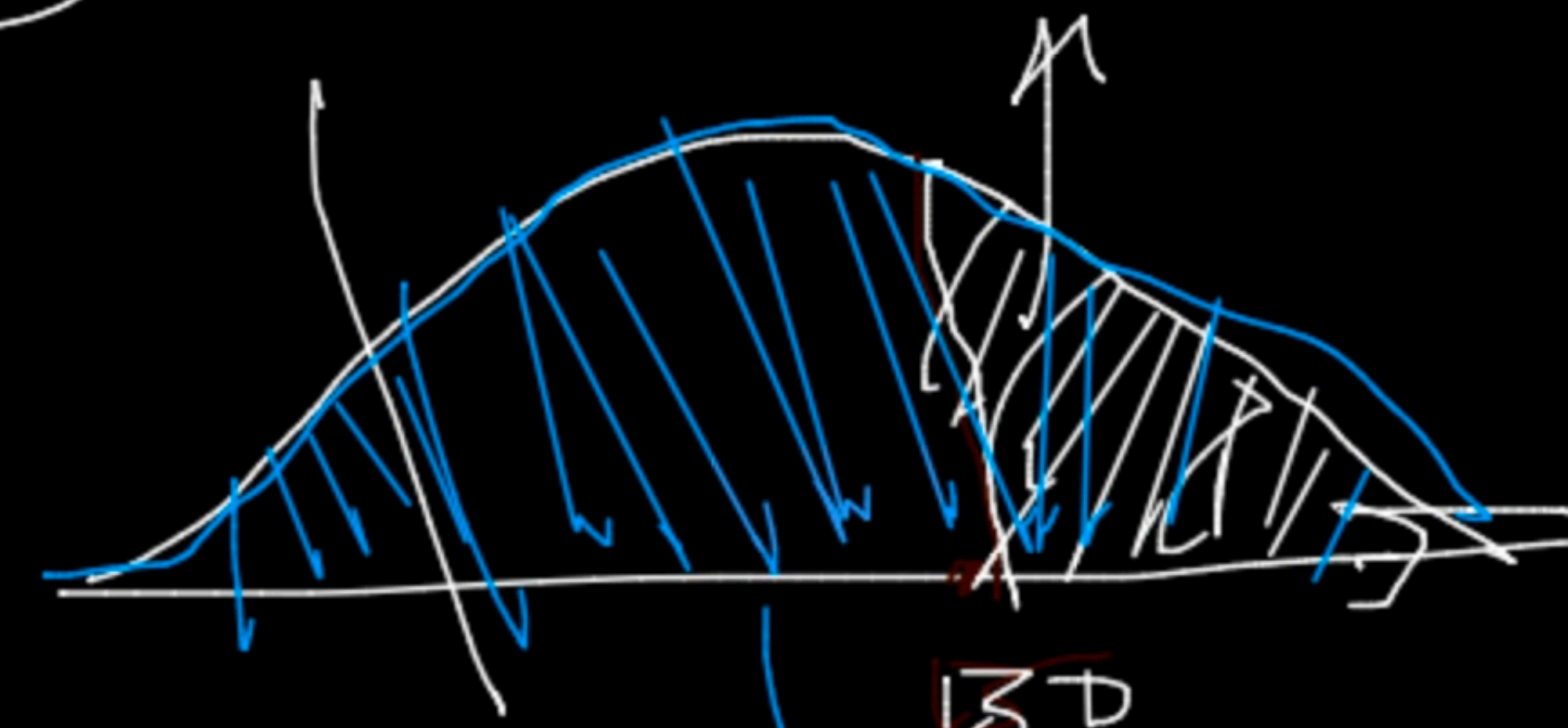
plotting



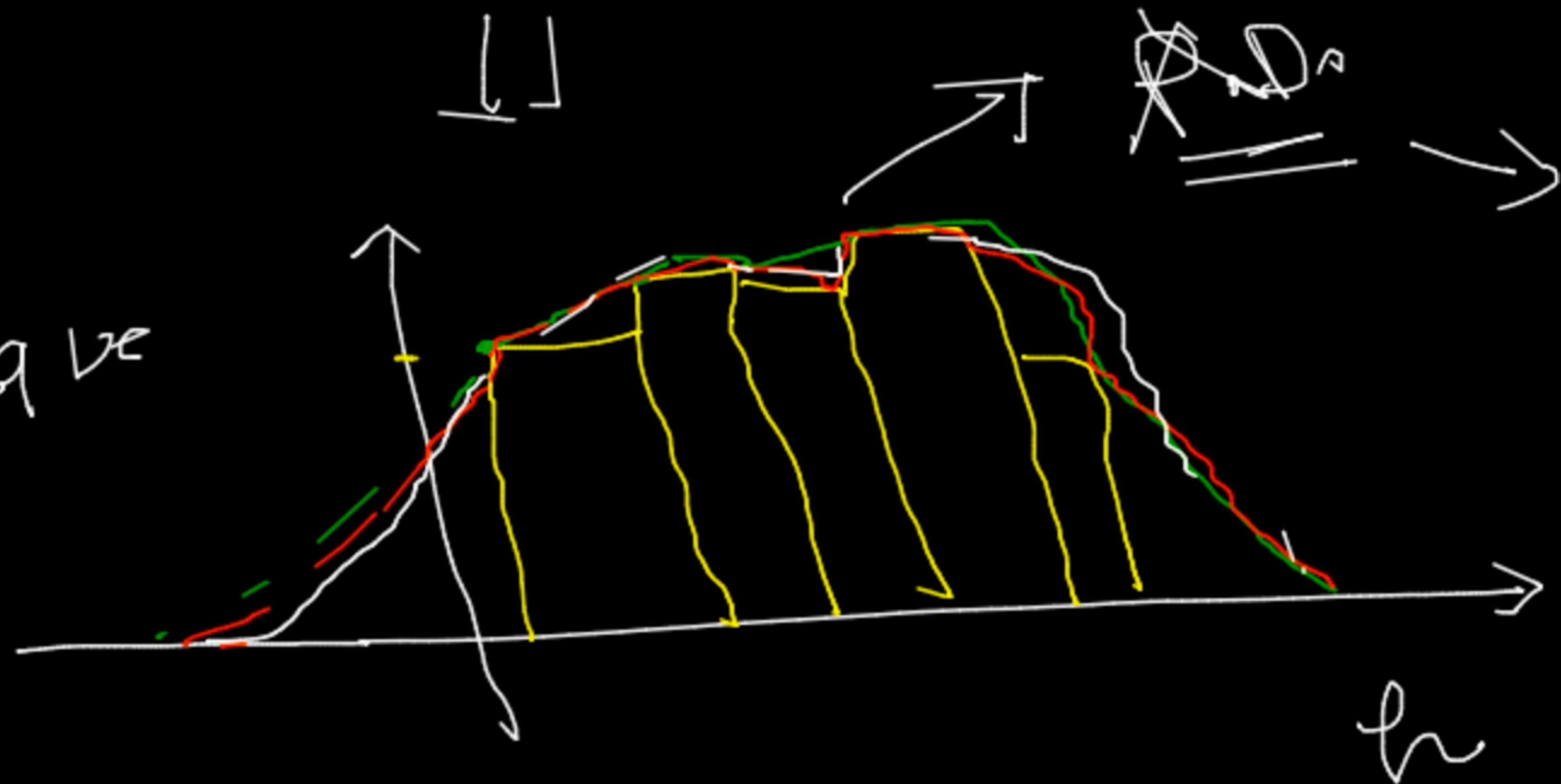
Binning

100 - 120	→ 50
120 - 140	→ 60
140 - 160	→ 60
160 - 180	
180 - ∞	

$\int_{130}^{\infty} f(x) dx$



frequency



~~POD~~

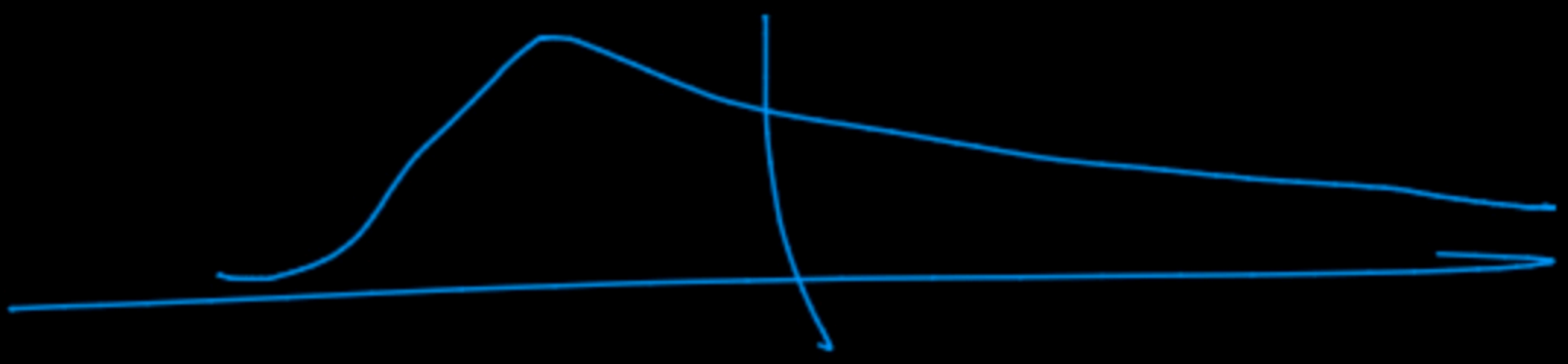
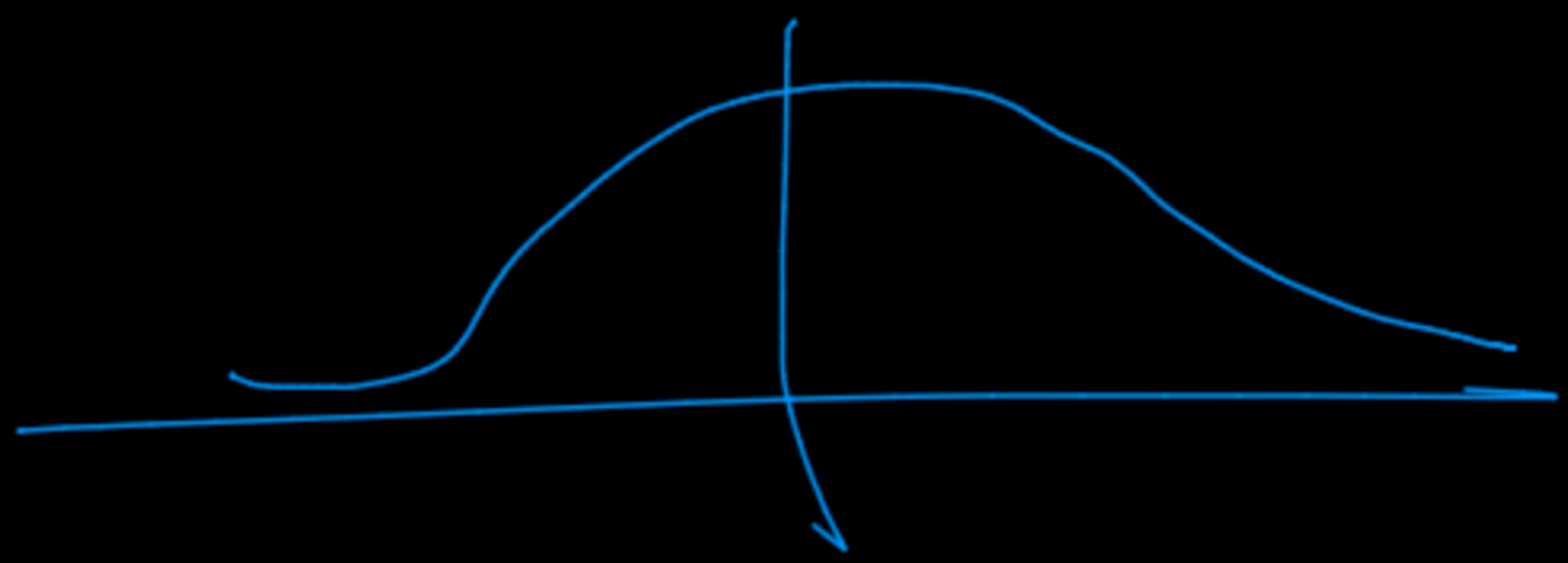
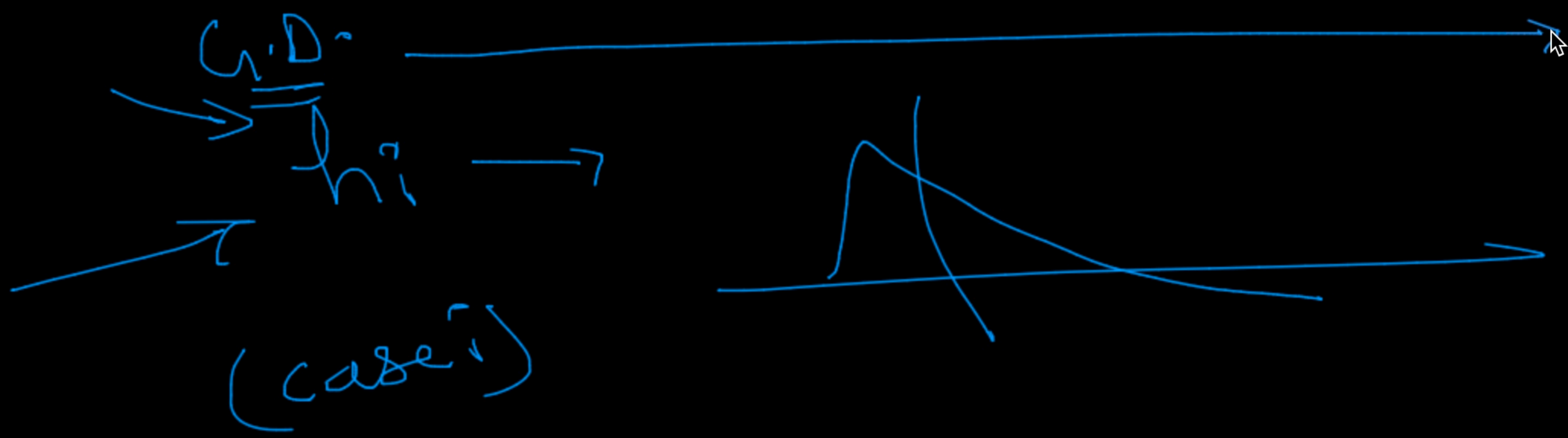
Distribution of Data

podis

Kernel

Add

$\int_{-\infty}^{\infty} f(x) dx$



$(\mu \text{ \& } \sigma)$

std. deviation

$$X \sim N(\mu, \sigma)$$

$\Rightarrow X$ as a random variable that follows Normal distribution with mean μ & std-dev

variance

$$X \sim N(\mu, \sigma^2)$$

$$P(X = 135)$$

$$P(X = 135)$$

$$= \frac{1}{\sqrt{2\pi\sigma}} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}$$

$$\sigma = 1 \quad \mu = 0$$

$$p(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x^2\right)$$

OR $\leadsto$

$$p(x) = e^{(-x)^2}$$

$\hookrightarrow \exp(-x^2)$

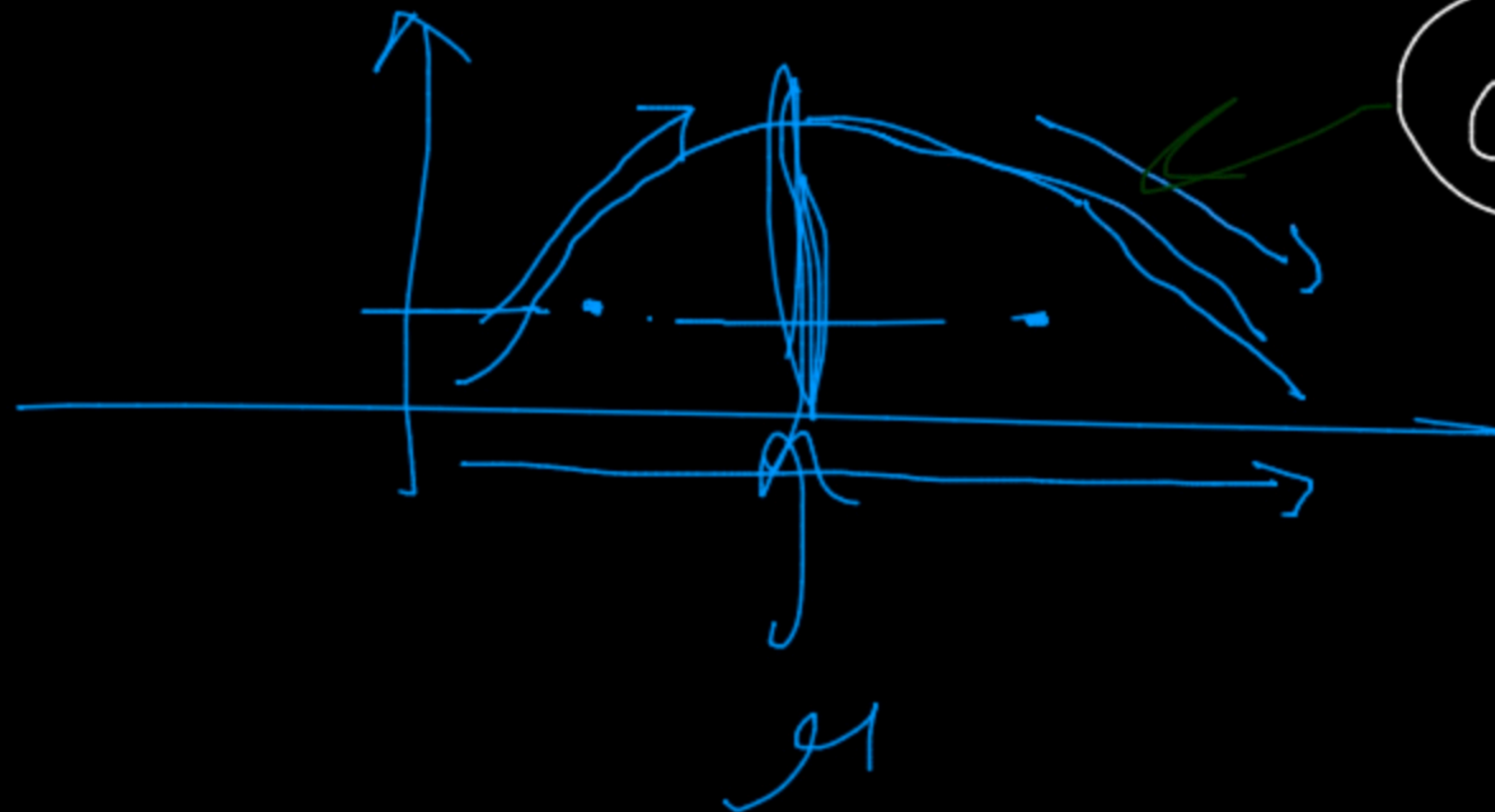
1) Symmetric $\rightarrow$ 2) as x moves away from 0

$$f = \exp(-x^2)$$

$\Rightarrow \log(f)$

$f \downarrow$

S.P



as $x \uparrow$
 $f \downarrow$
 But

C.D.F.

eg.

~~122~~

122

154

156

158

$P(x)$

0.11

0.9

0.71

0.12

0.11

0.20

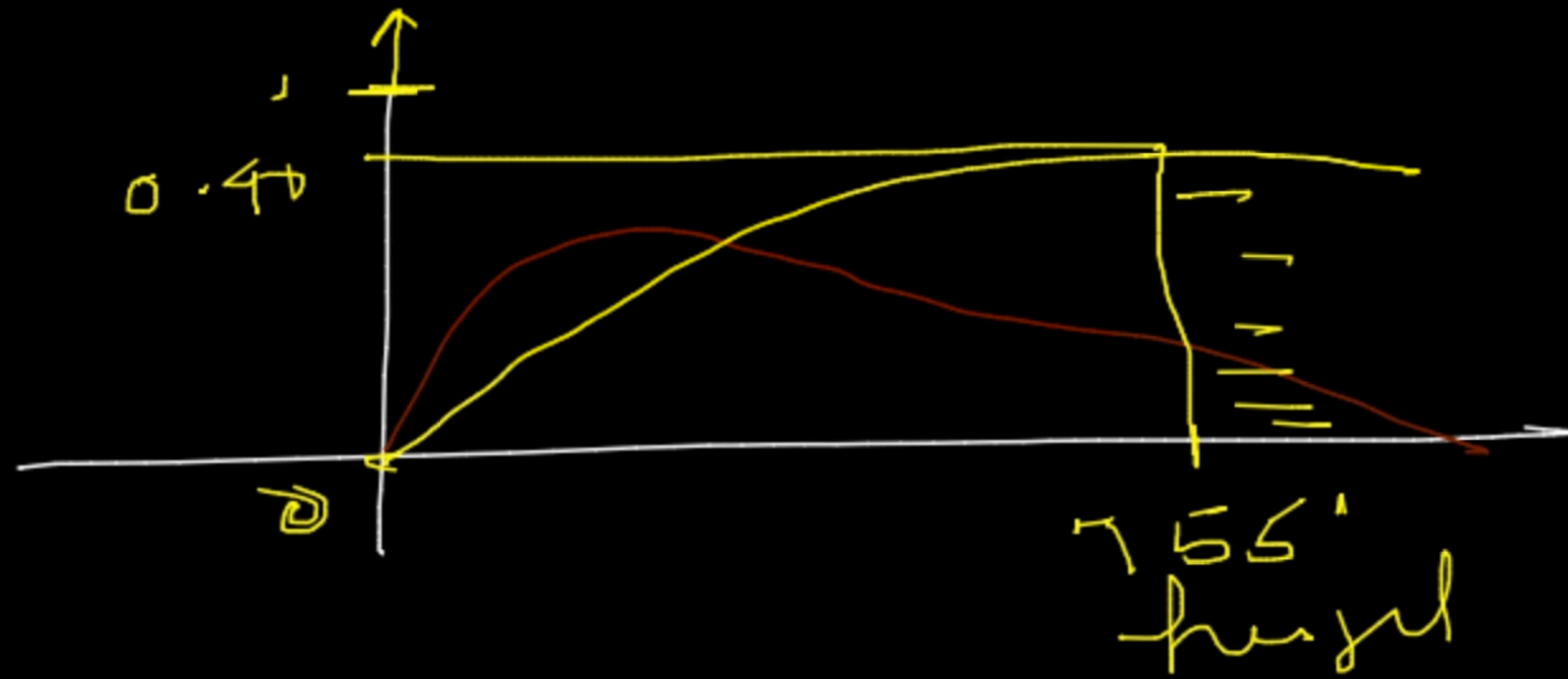
0.31

0.43

1.1

using the formula

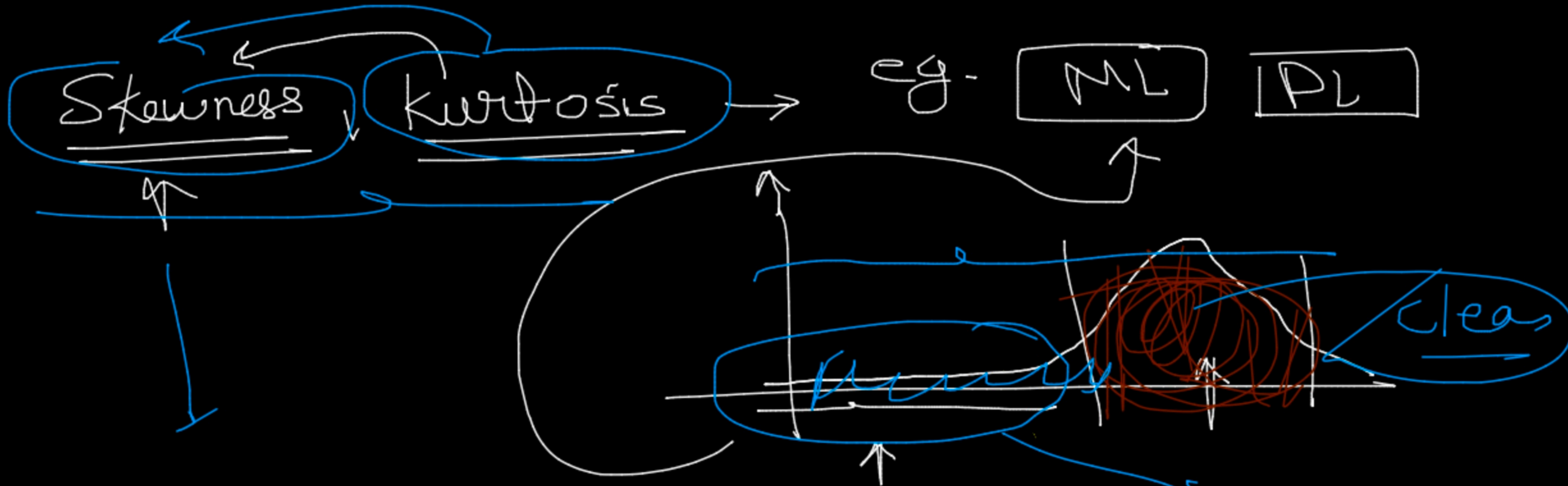
Plot



$x \rightarrow 1.1$

1.1

68-95-99 Rule



i) Negative Skewness

ii) Positive Skewness

left Skewness

n